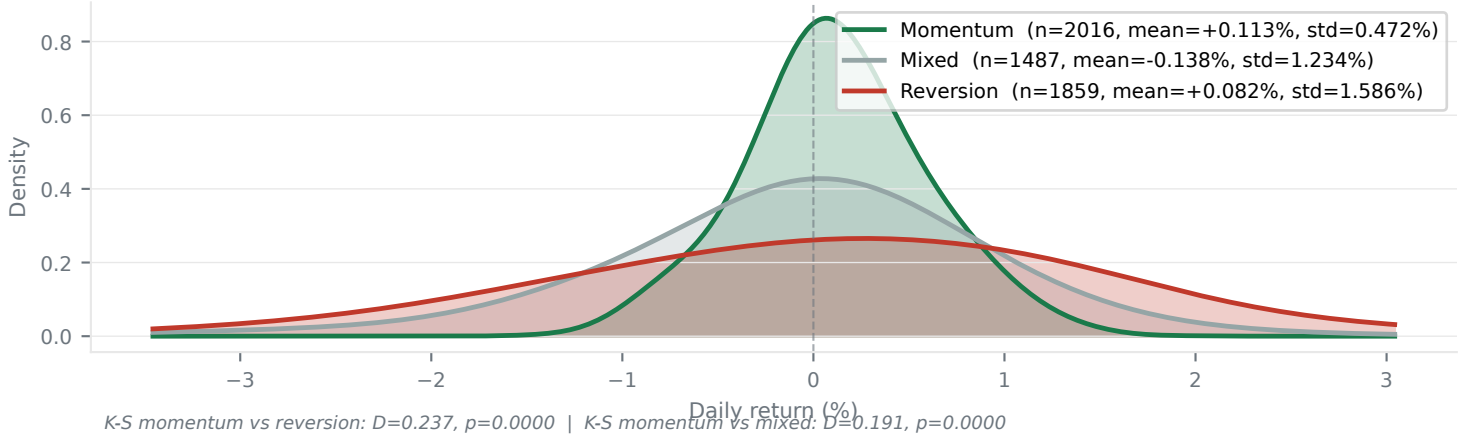
# Regime Ensemble - Technical Report

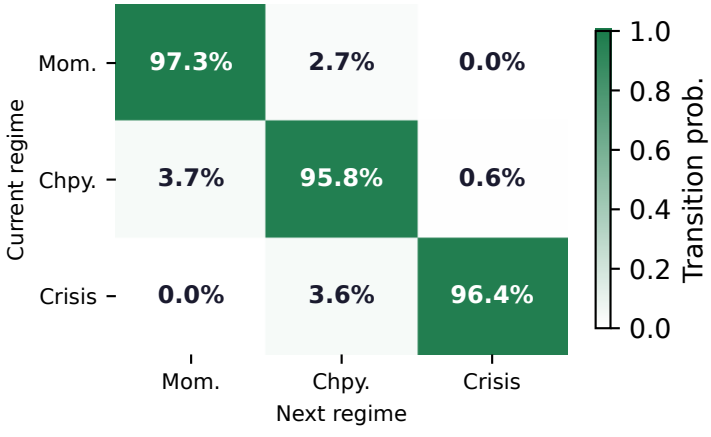


## Return Distributions by Regime (Kernel Density Estimate)



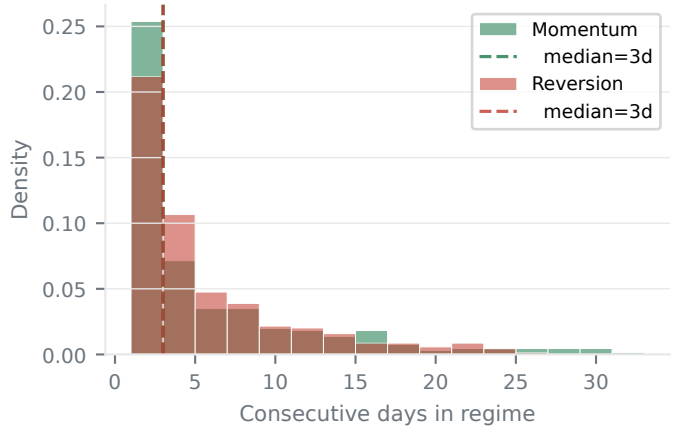

*K-S momentum vs reversion: D=0.237, p=0.0000 | K-S momentum vs mixed: D=0.191, p=0.0000*

## Markov Transition Matrix



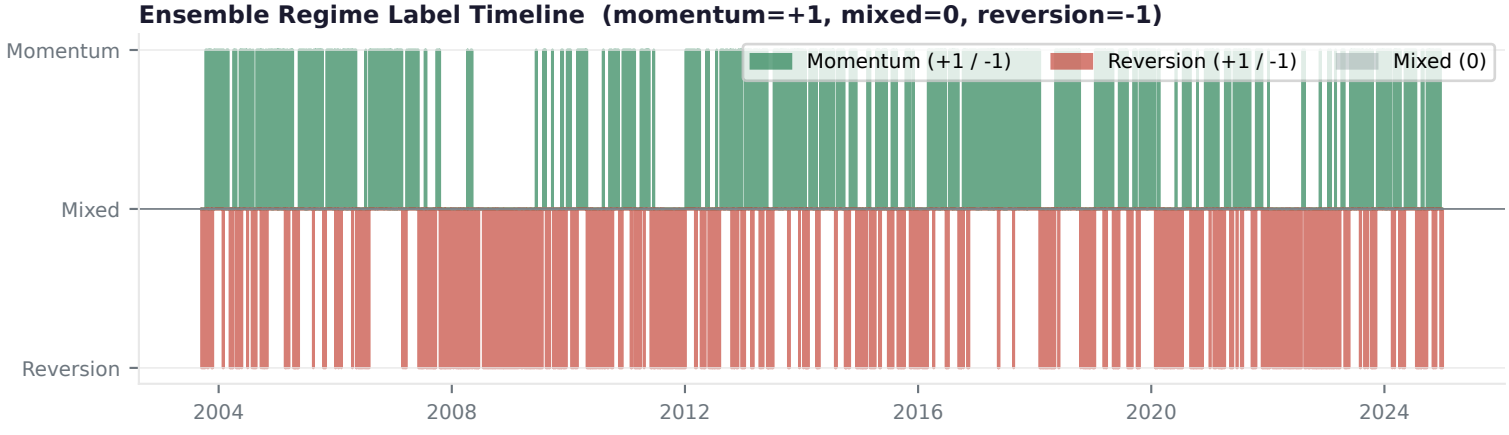

*Diagonal = regime persistence. High diagonal = sticky regimes.*

## Regime Sojourn Time Distribution




*Short sojourn -> fast flip -> higher turnover -> cost drag.*

## Ensemble Regime Label Timeline (momentum=+1, mixed=0, reversion=-1)



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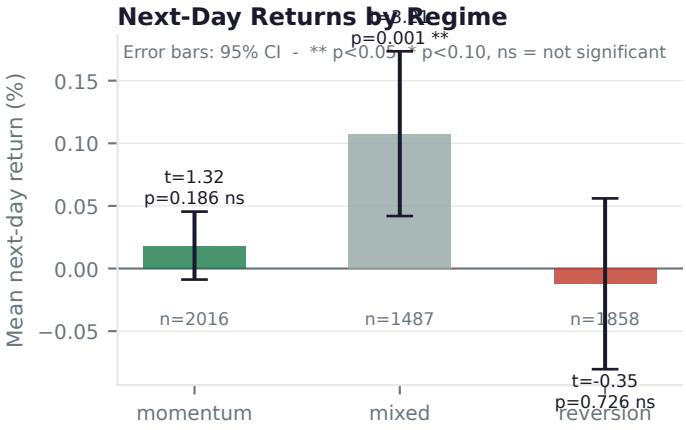


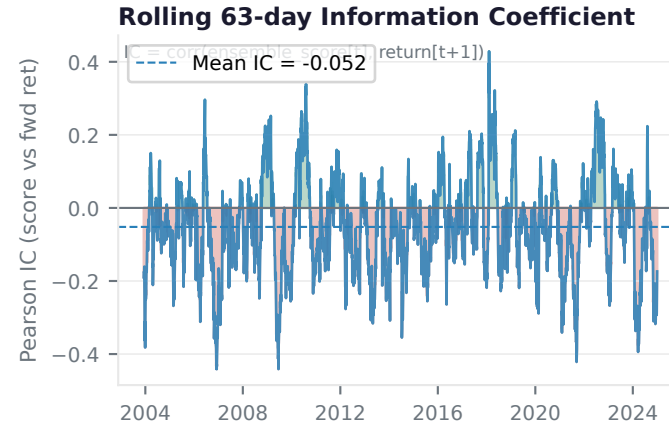


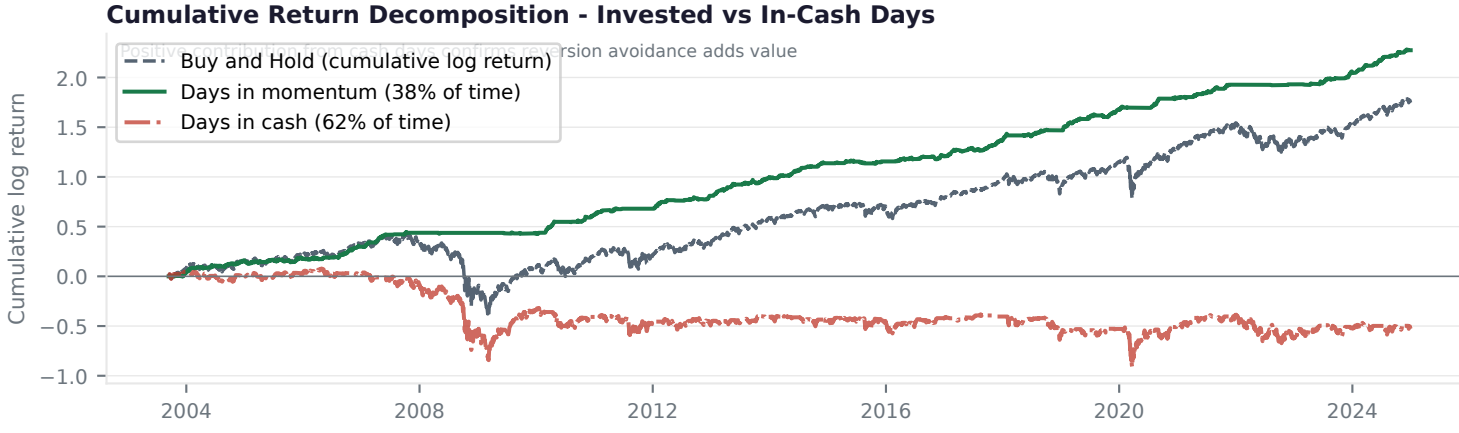


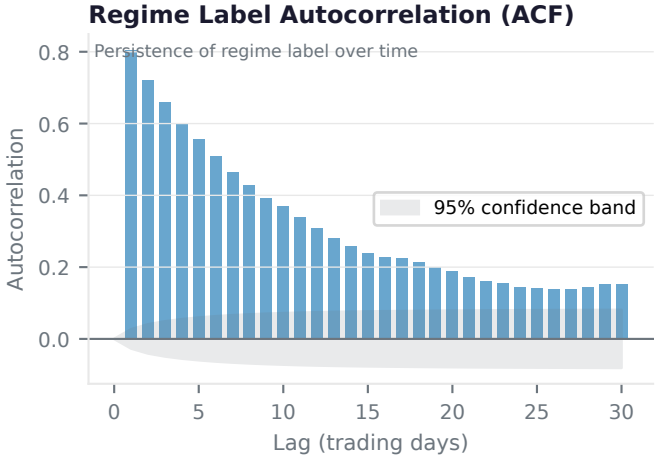


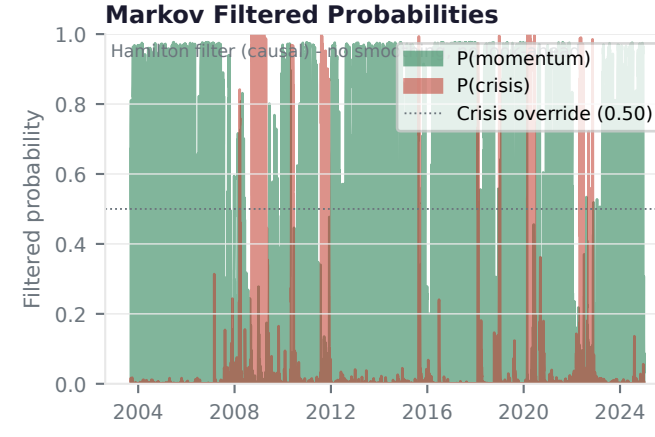

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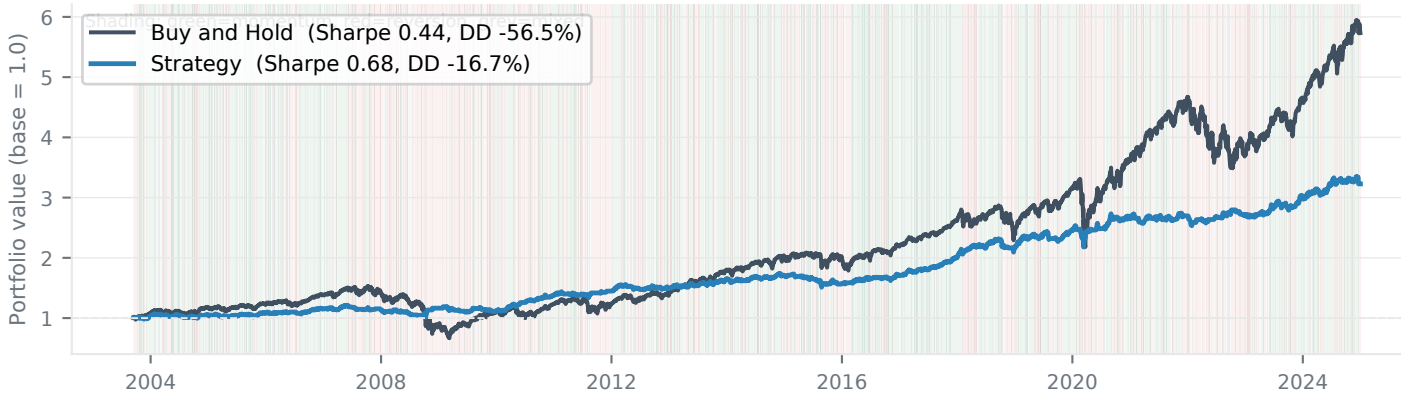
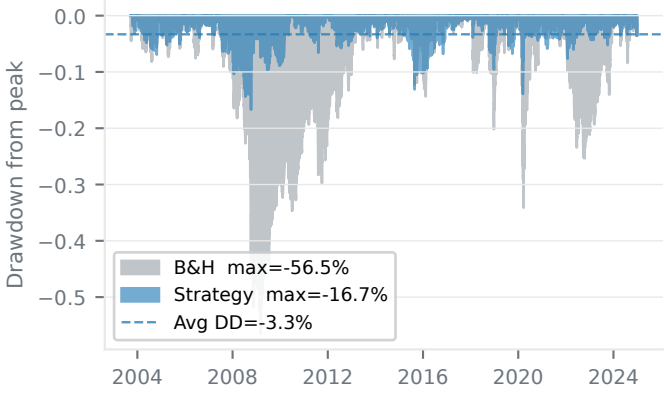
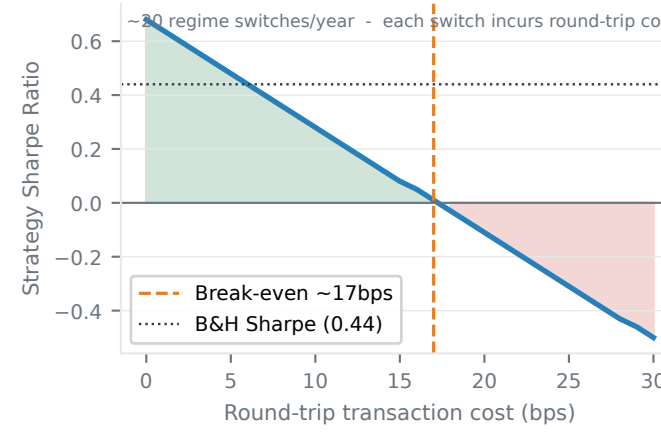
## Equity Curves - Regime-Shaded Background (0 bps, 1-day lag)




## Underwater Drawdown




## Sharpe vs Transaction Cost (0-30 bps)




## Full Performance Attribution (SPY 2000-2025, 0 bps)

| Metric | Strategy | Buy and Hold |
|---|---|---|
| CAGR | **+5.3%** | +6.7% |
| Vol (ann.) | **8.1%** | 18.7% |
| Sharpe | **0.65** | 0.36 |
| Sortino | **0.69** | 0.43 |
| Calmar | **0.31** | 0.11 |
| Max DD | **-17.1%** | -60.7% |
| Win rate | **36.2%** | 54.7% |
| Profit fctr | **0.93** | 0.90 |
| Information Ratio | **-0.18** | n/a |
| Time Invested | **37.6%** | 100.0% |